

逻辑函数的代数化简法

试化简以下逻辑函数：

$$F_1 = \overline{A}\overline{B}\overline{C}D + A\overline{B}\overline{C}D$$

$$F_2 = A\overline{C} + A\overline{B}\overline{C}(E + G)$$

$$F_3 = A\overline{C} + \overline{A}D + CD$$

$$F_4 = \overline{A}\overline{B}\overline{C} + \overline{B}E + CE + \overline{A}DE$$

$$F_5 = \overline{A}\overline{B} + (AB + A\overline{B} + \overline{A}B)C$$

$$F_6 = A(A + \overline{B} + \overline{C})(\overline{A} + C + D)(E + \overline{C}\overline{D})$$

$$F_7 = AB + A\overline{C} + \overline{B}C + B\overline{C} + \overline{B}D + B\overline{D} + ADE(F + G)$$

$$\begin{aligned}F_1 &= \overline{A}\overline{B}\overline{C}D + A\overline{B}\overline{C}D \\&= AD(\overline{B}\overline{C} + B\overline{C}) \\&= AD\end{aligned}$$

吸收律+代入规则

$$\begin{aligned}F_2 &= \overline{A}\overline{C} + A\overline{B}\overline{C}(E + G) \\&= \overline{A}\overline{C}\end{aligned}$$

$$F_3 = \overline{A}\overline{C} + \overline{A}\overline{D} + CD$$

$$= \overline{A}\overline{C} + (\overline{A} + C)D$$

$$= \overline{A}\overline{C} + \overline{\overline{A}CD}$$

$$= \overline{A}\overline{C} + D$$

配项法+代入规则

$$F_5 = \overline{\overline{A}\overline{B}} + (AB + \overline{A}\overline{B} + \overline{A}B)C$$

$$= \overline{\overline{A}\overline{B}} + (A + B)C$$

$$= \overline{\overline{A}\overline{B}} + \overline{\overline{A}BC}$$

$$= \overline{\overline{A}\overline{B}} + C$$

恒等式

$$F_4 = \overline{A}\overline{B}\overline{C} + \overline{B}E + CE + \overline{A}DE$$

$$= \overline{A}\overline{B}\overline{C} + (\overline{B} + C)E + \overline{A}DE$$

$$= \overline{A}\overline{B}\overline{C} + \overline{BCE} + \overline{A}DE$$

反演律+配项法

$$= \overline{A}\overline{B}\overline{C} + \overline{BCE} + \overline{A}E + \overline{A}DE$$

$$= \overline{A}\overline{B}\overline{C} + \overline{BCE} + \overline{A}E$$

$$= \overline{A}\overline{B}\overline{C} + \overline{BCE}$$

$$F_6 = A(A + \bar{B} + \bar{C})(\bar{A} + C + D)(E + \bar{C}\bar{D})$$

$$\therefore F_6' = A + \bar{A}\bar{B}\bar{C} + \bar{A}CD + E(\bar{C} + \bar{D})$$

$$= A + \bar{A}CD + E(\bar{C} + \bar{D})$$

$$= A + CD + E(\bar{C} + \bar{D})$$

$$= A + \underline{CD} + \bar{C}E + \bar{D}E$$

$$= A + CD + \bar{C}E + \boxed{DE} + \bar{D}E$$

$$= A + CD + E$$

$$\therefore F_6 = A(C + D)E$$

$$\begin{aligned}
F_7 &= \underline{AB} + A\bar{C} + \underline{\bar{B}C} + B\bar{C} + \bar{B}D + B\bar{D} + ADE(F + G) \\
&= AB + A\bar{C} + \boxed{AC} + \bar{B}C + B\bar{C} + \bar{B}D + B\bar{D} + ADE(F + G) \\
&= \underline{AB} + A + \bar{B}C + B\bar{C} + \bar{B}D + B\bar{D} + \underline{ADE(F + G)} \\
&= A + \underline{\bar{B}C} + B\bar{C} + \bar{B}D + \underline{B\bar{D}} \\
&= A + \bar{B}C + \boxed{B\bar{D}} + \boxed{\bar{C}\bar{D}} + B\bar{C} + \bar{B}D \\
&= A + \boxed{\bar{B}C} + B\bar{C} + \underline{\bar{B}D + \bar{C}\bar{D}} \\
&= A + \bar{B}C + \bar{B}D + \bar{C}\bar{D} \\
& (= A + \bar{B}C + B\bar{C} + \bar{B}D + B\bar{D} + \boxed{\bar{C}\bar{D}}) \\
&= A + \bar{B}C + \boxed{B\bar{C}} + \underline{B\bar{D} + \bar{C}\bar{D}} \\
&= A + \bar{B}C + \bar{B}D + \bar{C}\bar{D}
\end{aligned}$$